Project Management and Professional Ethics

MGT 401

Identifying, Forecasting, and Comparing Costs and Benefits

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Forecasting

- Predict key variables for proposal development.
- Use **What-if analysis**, but don't rely on it alone.

- **No Historical Data: Judgment Methods**: Sales force estimates, surveys, scenarios, historical analogies.
- **Historical Data Available: Conditional Forecasting** (Causal relationships): Correlation, regression, econometrics, leading indicators, input/output models.
- **Unconditional Forecasting** (No causal relationships): Graphical judgment, moving averages, time-series analysis.
    - Benefits: Simple, reliable, cost-effective.

Estimation of Trends

- Moving average estimates trends by calculating the mean of data over a fixed period.
- Example: A three-month moving average uses sales data from January, February, and March to predict April.
- smoothing data but has limitations in handling extreme fluctuations.
- Trend forecasting is crucial for justifying projects with reliable information.
- Smooths out seasonal, cyclical, and random patterns, making trends clearer.
- More affected by extreme values compared to methods like graphical judgment or least squares.
- Useful for smoothing data but has limitations in handling extreme fluctuations.
- Trend forecasting is crucial for justifying projects with reliable information.

Identifying Benefits and Costs

Intangible Benefits (Difficult to Measure):

- Improved decision-making.
- Enhanced accuracy.
- Better customer service and competitiveness.
- Positive business image.
- Increased job satisfaction by reducing tedious tasks.

Identifying Benefits and Costs

Tangible Benefits (Measurable in Money):

- Faster processing speed.
- Access to otherwise inaccessible or timely information.
- Reduced employee time for specific tasks.
- Superior calculating power.
- Measurable in dollars, resources, or time saved.

Identifying Benefits and Costs

Tangible Costs (Easily Measurable):

- Equipment (computers, terminals).
- Resource costs.
- Salaries of systems analysts, programmers, and other employees.
- Requires cash outlay and can be accurately projected.

Identifying Benefits and Costs

Intangible Costs (Difficult to Measure):

- Loss of competitive edge.
- Decline in company image or reputation.
- Increased customer dissatisfaction.
- Ineffective decision making due to inaccessible or untimely information.
- Hard to quantify but must be considered in decision-making.

Comparing Costs and Benefits

- Break-even Analysis
- Payback
- Cashflow analysis
- Present value analysis

Break-even Analysis

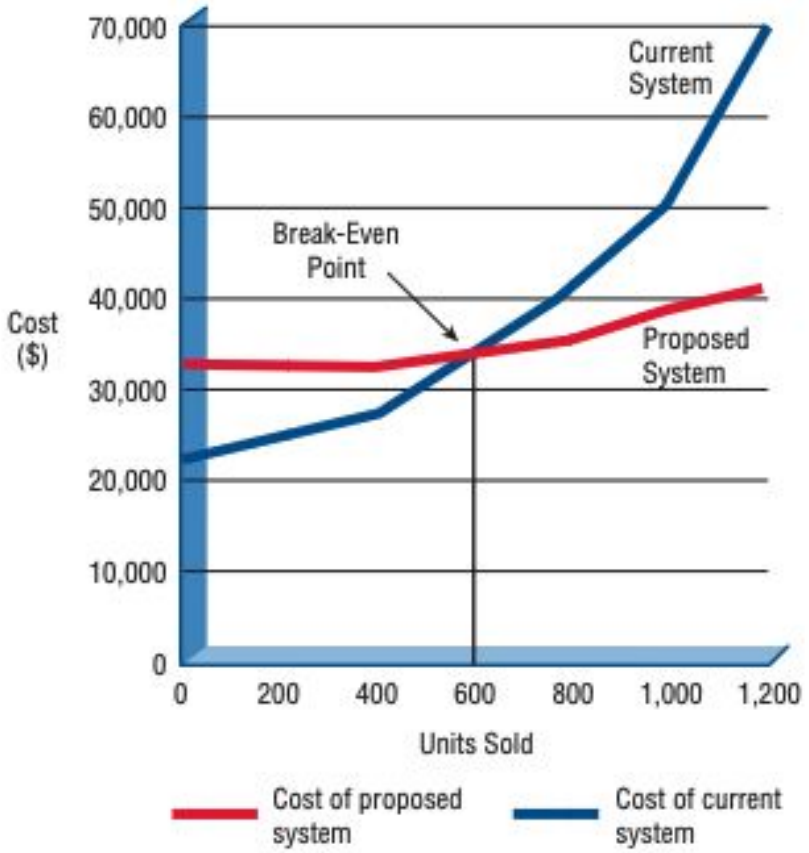
Purpose: Determines the point where the proposed system becomes profitable compared to the current system.

Break-Even Point: The point where the total costs of the current system and proposed system intersect.

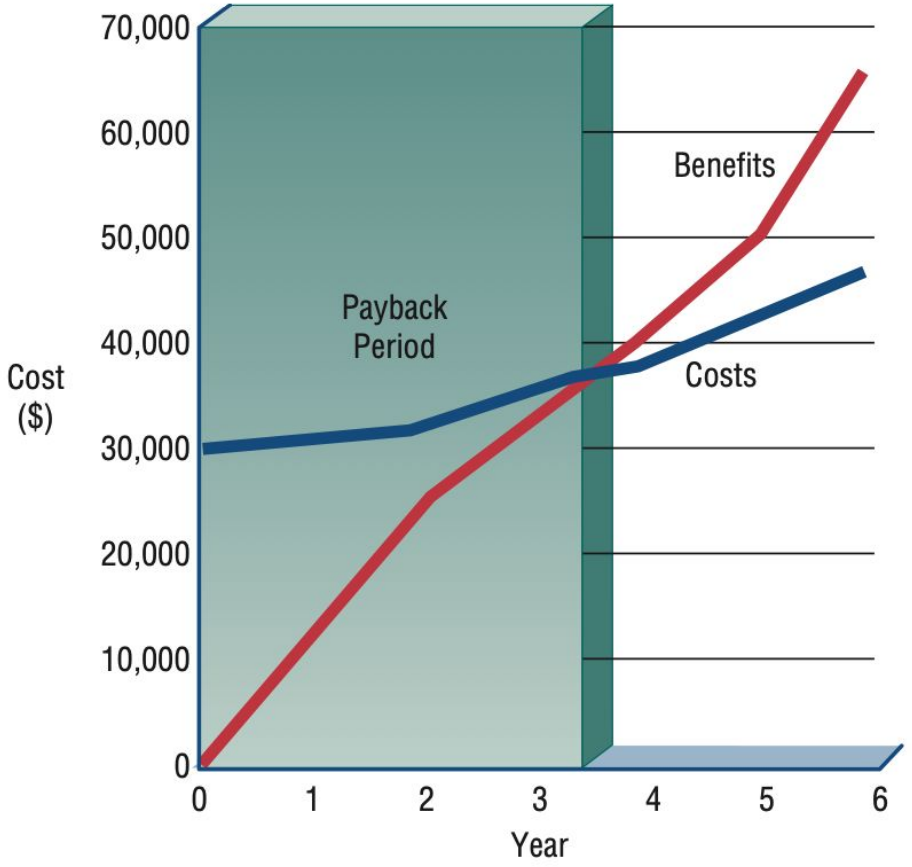
Costs Considered:

- **Total Costs**: Recurring operational costs + one-time developmental costs (tangible costs).
- **Current System**: Costs rise with increasing volume.
- **Proposed System**: High initial costs, but lower incremental costs for higher volume.

Break-even Analysis



Break-even Analysis



Cumulative benefits from proposed system

Cumulative costs of proposed system

Year	Cost ($)	Cumulative Costs ($)	Benefits ($)	Cumulative Benefits ($)
0	30,000	30,000	0	0
1	1,000	31,000	12,000	12,000
2	2,000	33,000	12,000	24,000
3	2,000	35,000	8,000	32,000
4	3,000	38,000	8,000	40,000
5	4,000	42,000	10,000	50,000
6	4,000	46,000	15,000	65,000

Break-even analysis can also determine how long it will take for the benefits of the system to pay back the costs of developing it.

Cash-Flow Analysis

- Examines the direction, size, and pattern of cash flow associated with the proposed system.
- If the new system replaces an old one and does not generate additional revenue, only cash outlays (equipment, salary) are considered.
- Justification cannot be based on new revenues; focus shifts to other tangible benefits.
- The system must be carefully evaluated for non-revenue benefits before further pursuit.

Cash-Flow Analysis

	Year 1				Year 2
	Quarter 1	Quarter 2	Quarter 3	Quarter 4	Quarter 1
Revenue	$5,000	$20,000	$24,960	$31,270	$39,020
Costs					
Software development	10,000	5,000			
Personnel	8,000	8,400	8,800	9,260	9,700
Training	3,000	6,000			
Equipment lease	4,000	4,000	4,000	4,000	4,000
Supplies	1,000	2,000	2,370	2,990	3,730
Maintenance	0	2,000	2,200	2,420	2,660
Total Costs	26,000	27,400	17,370	18,670	20,090
Cash Flow	–21,000	–7,400	7,590	12,600	18,930
Cumulative Cash Flow	–21,000	–28,400	–20,810	–8,210	10,720

Present Value Analysis

- Present value analysis helps a systems analyst present to business decision makers the time value of the investment in the information system as well as the cash flow.
- Present value is a way to assess all the economic outlays and revenues of the information system over its economic life and to compare costs today with future costs and today's benefits with future benefits.

Formula: PV = $\frac{1}{(1+r)^n}$

where:

r = discount rate (cost of money), *n* = number of periods

Present Value Analysis

	Year						
	1	2	3	4	5	6	Total
Costs	$40,000	42,000	44,100	46,300	48,600	51,000	272,000
Benefits	$25,000	31,200	39,000	48,700	60,800	76,000	280,700

	Year						
	1	2	3	4	5	6	Total
Costs	$40,000	42,000	44,100	46,300	48,600	51,000	
Multiplier	.89	.80	.71	.64	.57	.51	
Present Value of Costs	35,600	33,600	31,311	29,632	27,702	26,010	183,855
Benefits	$25,000	31,200	39,000	48,700	60,800	76,000	
Multiplier	.89	.80	.71	.64	.57	.51	
Present Value of Benefits	22,250	24,960	27,690	31,168	34,656	38,760	179,484

Guidelines for Analysis

1. Use break-even analysis if the project needs to be justified in terms of cost, not benefits, or if benefits do not substantially improve with the proposed system.
2. Use payback when the improved tangible benefits form a convincing argument for the proposed system.
3. Use cash-flow analysis when the project is expensive relative to the size of the company or when the business would be significantly affected by a large drain (even if temporary) on funds.
4. Use present value analysis when the payback period is long or when the cost of borrowing money is high.

Project Scheduling

- **Planning**: Form a systems analysis team, assign tasks, estimate durations, and schedule activities.
- **Control**: Monitor progress, compare actual performance with the plan, and adjust as necessary.
- **Task Breakdown**: Decompose major activities (analysis, design, implementation) into detailed subtasks.
- **Time Estimates**: Use most likely, pessimistic, and optimistic estimates with a weighted average to project activity durations. Structured activity descriptions help minimize surprises.

Two new techniques:

- **Gantt Chart**: Visual tool for scheduling tasks and tracking progress.
- **PERT Diagram**: Network diagram to map out task sequences and identify the critical path.